

Core Software Components

- **Primary Processing Languages:**
 - **Python 3:** Deployed on the Raspberry Pi for high-level tasks including image processing, sensor integration, state management, and steering calculation.
 - **C++ (Embedded):** Deployed on the Arduino Nano for deterministic, real-time control of motor PWM and servo steering commands.
- **Core Libraries & Dependencies:**
 - **Raspberry Pi Environment:** [Picamera2](#) / [libcamera](#) (high-frame-rate hardware camera pipeline), [OpenCV](#) & [NumPy](#) (vectorized color extraction and spatial matrix math), [pyserial](#) (inter-board communication), [time](#).
 - **Arduino Environment:** Built-in hardware [Serial](#) interface, custom low-level motor driver logic.
- **Functional Task Distribution:**

